Indian Institute of Technology Kharagpur
Department of Industrial and Systems Engineering
BTP-I Mid-Semester Project Summary
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LLM and Agentic AI-Based Decision-Making for Autonomous Robotic Systems

Background and Motivation

Autonomous robotic systems such as unmanned aerial vehicles (UAVs), mobile robots, and robotic manipulators are increasingly expected to operate in dynamic environments while satisfying operational, physical, and resource constraints. Conventional path-planning and optimization techniques are effective when the objective, environment, and constraints are clearly defined. However, higher-level autonomous operation may additionally require interpreting mission objectives, selecting appropriate planning methods, adapting decisions when conditions change, reasoning about available capabilities, and recovering from unsuccessful actions.

Recent advances in Large Language Models (LLMs) and agentic AI provide a possible mechanism for introducing such higher-level reasoning. The objective of this project is not to replace numerical optimization, path-planning algorithms, or low-level robot controllers with an LLM. Instead, the project investigates how LLM-based reasoning can coordinate specialized planning and control components, select suitable tools, interpret changing requirements, and make higher-level decisions while conventional algorithms continue to perform computational planning and control.

Thus, the problem can be viewed as a hierarchical decision-making system involving reasoning, tool and capability selection, planning, execution, feedback, and replanning.

Related Work and Research Direction

Recent research demonstrates several ways in which language models can contribute to robotic planning. *Agentic-Nav* proposes a hierarchical multi-agent architecture in which an LLM coordinates specialized agents responsible for perception and task execution, while lower-level trajectory planning remains separated from strategic reasoning [1]. Jiang *et al.* compare SLM-, LLM-, and agentic AI-based approaches for UAV-enabled wireless power transfer and introduce an agentic architecture based on Initializer, Actor, Critic, and Reflector agents for iterative path and energy optimization [2].

LLM-PD combines LLM-based semantic reasoning with multi-agent reinforcement learning for multi-UAV path planning. An LLM-based teacher generates coordinated expert strategies that are transferred to a smaller execution policy through policy distillation, imitation learning, and reinforcement learning [3]. Xiao *et al.* study LLM-based UAV swarm path planning for industrial systems using generated reasoning samples together with scene-memory and replay mechanisms for adaptation in dynamic environments [4].

These approaches indicate that LLMs can play different roles, including high-level reasoning, iterative optimization, expert-policy generation, and multi-agent orchestration. This motivates studying a more general architecture in which agentic reasoning can interact with conventional planning and optimization tools while remaining separated from the specific robotic platform used for execution.

Aim and Objectives

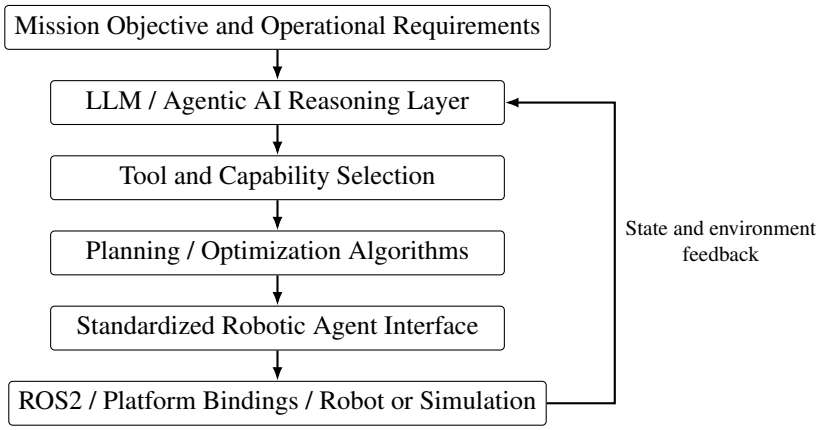
The aim of this project is to investigate how LLMs and agentic AI can be integrated with conventional planning, optimization, and robotic-control methods to provide adaptable autonomous decision-making for robotic systems.

The main objectives are:

- Study existing LLM-, SLM-, and agentic AI-based approaches for robotic decision-making and path planning.
- Develop a reasoning architecture in which an LLM-based system can interpret a mission, understand available robot functionality, and select or invoke suitable planning tools.
- Integrate conventional path-planning and optimization algorithms with the reasoning layer rather than replacing them with an LLM.
- Provide a common experimental interface so that planning and reasoning approaches can be evaluated across different robotic systems.
- Evaluate the system in simulation using measures related to planning quality, efficiency, adaptability, constraint satisfaction, and decision-making overhead.

Proposed System Architecture

The proposed architecture separates high-level reasoning from numerical planning and low-level robotic control.



The LLM or agentic system acts primarily as a reasoning and orchestration component. Depending on the task, it may select an appropriate planning algorithm, determine parameters and constraints, invoke tools, evaluate results, and request replanning when necessary. Agentic orchestration frameworks such as LangGraph or LangChain can be used to implement such reasoning loops.

Specialized algorithms such as graph search, sampling-based planning, mathematical optimization, or learned policies remain responsible for computational planning. Similarly, stabilization, attitude control, actuator control, and other fast control loops remain the responsibility of established robotic controllers such as PX4.

Current Progress

A major part of the work completed so far has focused on developing the experimental infrastructure required to study the proposed architecture systematically. Different robotic platforms expose different controllers, sensors, communication interfaces, and operations. Directly connecting each planning or agentic algorithm to every platform would repeatedly introduce platform-specific implementation into the experiments.

To address this problem, an initial version of a general robotic *Agent framework* has been developed. A robotic system is represented using a structured description of its available *Properties*, such as state information, and *Capabilities*, representing operations that the system can perform. Platform-specific bindings translate these generic concepts to the underlying robotic system, while a generated Python API provides the interface used by applications and planning algorithms.

The current prototype includes:

- a technology-independent Agent representation for Properties, Capabilities, communication Channels, and operational Constraints;
- resolution of developer-defined Agents into ROS2 realizations and generated Python interfaces;
- a modular PX4 binding supporting UAV state information and flight-control operations;
- support for multiple runtime Agent instances and instance-specific configuration;
- sensor integration through capability-based streaming interfaces; and

- validation using ROS2, PX4 Software-In-The-Loop (SITL), and simulated sensor streams.

This framework is not the final research objective. It provides the experimental layer required to evaluate different planning algorithms, reasoning models, and robotic configurations without rewriting robot-specific communication infrastructure for every experiment. A separate consumer application is also being used to validate that planning algorithms can interact with a simulated PX4 UAV through the generated interface without directly depending on PX4 or ROS2 communication details.

The current prototype implementation is available at:

github.com/preethbera/robot-agent-framework

Planned Experimental Study and Further Development

The next phase will proceed along two connected directions: experimental investigation of LLM/agentic decision-making and continued development of the experimental infrastructure.

Initially, simple planning experiments will be performed in which a mission objective and a set of planning tools are available to the system. These may include geometric methods, graph-search algorithms, sampling-based planners, optimization procedures, or other specialized approaches. The reasoning layer will determine which tools and robot capabilities should be used, while the resulting motion commands will be executed through the standardized Agent interface.

Experiments will then be extended to scenarios involving dynamic obstacles, changing mission requirements, multiple candidate planning methods, failure of an initially selected strategy, replanning, and eventually heterogeneous or multi-robot systems. This will allow comparison between conventional fixed planning pipelines and systems in which an LLM or agentic architecture performs higher-level reasoning, tool selection, and adaptation.

In parallel, the Agent framework will continue to be developed as new experimental requirements emerge. The current implementation is an initial prototype establishing the core representation, binding, communication, API generation, multi-Agent, and sensor mechanisms. Future experiments may require support for additional robotic platforms, sensor types, capabilities, deployment configurations, or application interfaces. These extensions will be introduced based on concrete experimental requirements rather than by hard-coding platform-specific functionality into the generic framework.

Evaluation will consider metrics such as mission completion rate, path length, planning and decision time, replanning frequency, constraint violations, computational and LLM inference overhead, correctness of tool selection, and robustness to environmental or mission changes. PX4 SITL and Gazebo will initially provide a repeatable UAV simulation environment before extending the experiments to more complex robotic configurations.

Expected Outcome

The expected outcome is an experimentally evaluated architecture for combining LLM-based reasoning with specialized planning, optimization, and robotic-control components. The study will investigate where language-model-based reasoning provides useful adaptability and decision support, where conventional methods remain preferable, and how both can operate together within a structured autonomous system.

The project is expected to produce both an experimental methodology for studying agentic AI in robotic decision-making and a reusable software environment for conducting such experiments across different robotic configurations.

References

[1] K. Samarathunga, R. Gurusinghe, K. Sivasothynathan, J. Mars, V. Logeeshan, and C. Wanigasekara, "AgenticNav: A Hierarchical Multi-Agentic System for LLM-Driven Autonomous Problem-Solving in Robotics," *IEEE Access*, vol. 14, pp. 89934–89951, 2026. doi: 10.1109/ACCESS.2026.3703330.

[2] F. Jiang, L. Dong, L. Mao, K. Wang, X. Wang, and A. Jamalipour, "SLM, LLM or Agentic AI? Toward Intelligent UAV-Enabled WPT Systems in Low-Altitude Economy Networks," *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 5267–5281, 2026. doi: 10.1109/JSAC.2026.3704332.

[3] L. Tang, J. He, X. Liao, Q. Liu, and Q. Chen, "LLM-PD: A Large Language Model-Driven Policy Distillation-Based Method for Multi-UAV Path Planning," *IEEE Internet of Things Journal*, vol. 13, no. 8, pp. 15466–15484, 2026. doi: 10.1109/JIOT.2026.3656708.

[4] W. Xiao, C. Shi, M. Chen, A. V. Vasilakos, M. Chen, and A. Farouk, "LLM-based UAV Path Planning for Autonomous and Adaptive Industry Systems," *ACM Transactions on Autonomous and Adaptive Systems*, 2025. doi: 10.1145/3774655.

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